ROBOTICS · EMBEDDED SYSTEMS · PROJECT DOCUMENTATION

Electromagnetic Robotic Arm

A 3-degree-of-freedom arm that picks up iron and steel parts with an electromagnet instead of a mechanical gripper. It can, for example, pull iron filings out of sand and drop them somewhere else. Built on an Arduino Uno with three MG996R servos, 3D-printed links, and an electromagnet switched through a relay.

3 DOF base rotation · elbow · wrist	**3 × MG996R** 11 kg·cm metal-gear servos	**~14 s** per programmed pick-and-place cycle
50 °/s speed limit in software (1° every 20 ms)	**0 parts** in the gripper that move: the magnetic field holds the part	**2 tests** iron filings out of a mixture · nail moved to a target spot

My role	**Electronics, wiring and Arduino programming:** the servo and relay circuits, the electromagnet power circuit, soldered wiring, and the motion firmware. Plus the independent 2026 review and v2 redesign
Author	Shree Vaishnavi M G
Context	Academic project, B.E. Electronics & Communication, Dayananda Sagar College of Engineering, Bengaluru (course 22IDT28)
Timeline	June – September 2023 · shown at the department's Project Open Day, 9 Sep 2023
Platform	Arduino Uno (ATmega328P, 16 MHz) · Arduino IDE · `Servo.h`
This document	The project as it was built in 2023, plus a 2026 review and an improved v2 design (Sections 8–9)

1 Summary

Separating iron filings or scrap from sand or other mixed material by hand is slow, tiring, and can cut your hands on sharp edges. This project automates it with a small robotic arm. The arm's gripper is an electromagnet, so it only picks up ferromagnetic material and leaves everything else behind.

WHAT WAS BUILT

- A 3D-printed arm with three servo-driven joints: base rotation, elbow and wrist.
- A hand-wound electromagnet (copper wire around an iron-nail core) on the end of the arm.
- A relay that switches the electromagnet's current, kept separate from the Arduino's 5 V logic and powered by a Li-ion pack with a battery management system (BMS).
- An Arduino sketch that moves each joint through a programmed pick-and-place path, one degree at a time.

WHAT IT SHOWED

- The arm picked iron filings out of a mixture of materials and set them down elsewhere.
- It picked up a small iron nail and dropped it at a chosen spot.
- Every joint repeatedly moved through its programmed angle, and the relay switched the electromagnet on and off reliably.
- The project was shown working at the department's Project Open Day.

Why an electromagnet? A mechanical gripper's grip depends on its shape and the motor's torque, and it has to be lined up with each object. An electromagnet holds anything ferromagnetic, whatever its shape, can be switched on and off in milliseconds, and its pulling force can be raised by increasing the current. That makes it suited to loose, irregular material like filings and scrap, which a two-finger gripper can't pick up at all.

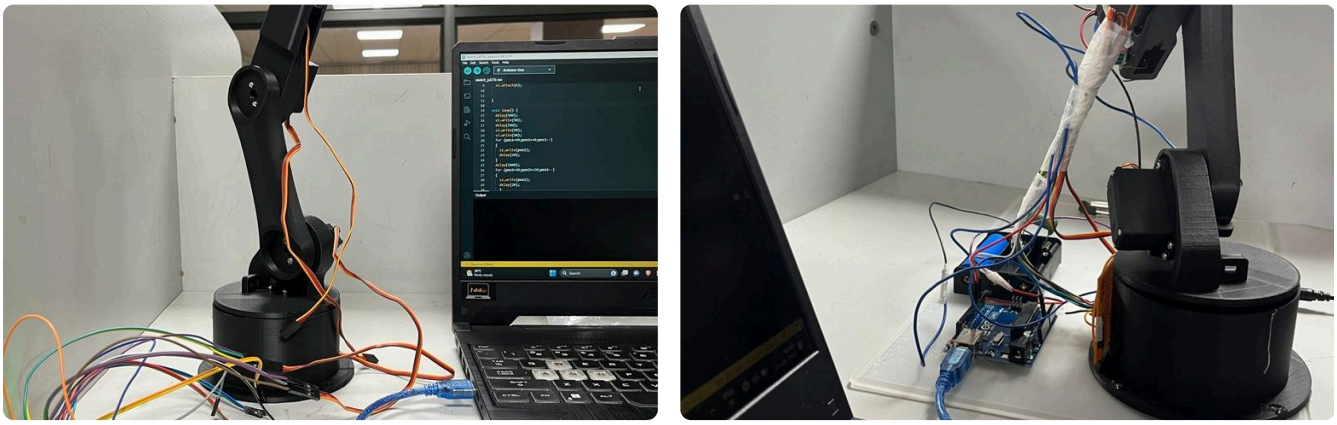


Figure 1. Left: the arm on the bench with the Arduino IDE. Right: the finished arm with the wound electromagnet on the end, ready for the pick-and-place tests.

2 System overview

The system has two separate parts: a **motion** part, where the Arduino generates PWM signals to drive three servos, and a **power** part, where a relay connects the battery to the electromagnet coil. Keeping the coil current away from the logic circuit is what makes the electromagnet safe to switch.

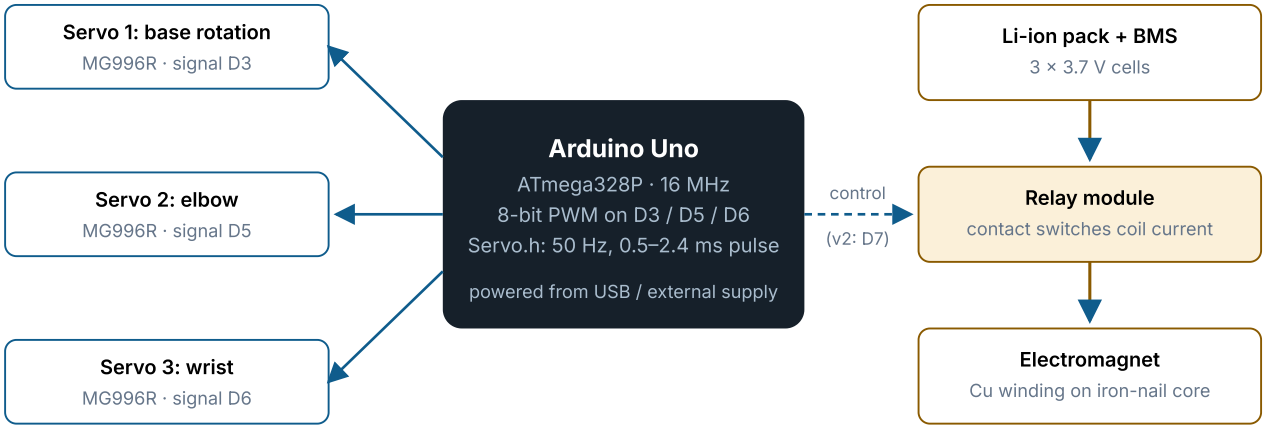


Figure 2. The two parts of the system: blue for logic and PWM, amber for the electromagnet's power. In the 2023 demo the relay was switched by hand; in the v2 design it is switched by the Arduino on D7 (Section 9).

Bill of materials

Part	Spec	Purpose
Arduino Uno R3	ATmega328P, 16 MHz, 32 KB flash, 2 KB SRAM, 6 PWM pins	Generates the motion sequence and the servo PWM signals
3 × MG996R servo	9.4 kg·cm @ 4.8 V, 11 kg·cm @ 6 V · 0.19 s/60° · 180° · 4.8–6 V	Base, elbow and wrist joints
3D-printed arm links	Based on an open-source desktop-arm model, printed and assembled	Base, links and the mount for the electromagnet
Electromagnet	Hand-wound copper wire on an iron-nail core	Picks up ferromagnetic material
Relay module	Single channel	Switches the coil current, which an Arduino pin (max 40 mA) cannot supply
3 × 3.7 V Li-ion cells + BMS	2.2 A rated, with a battery management system	Powers the electromagnet
Jumper wires, soldered joints	Servo leads: brown = GND, red = V+, orange = PWM	Wiring

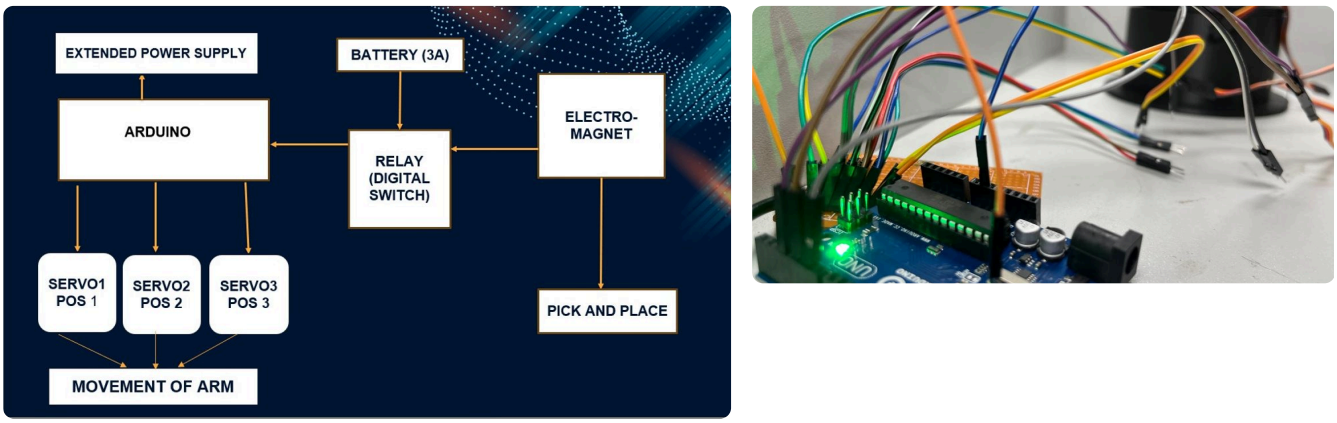


Figure 3. Left: the original 2023 block diagram. Right: the Arduino Uno with the three servo signal leads on D3, D5 and D6.

3 Mechanical design

The arm is a serial chain of three rotating joints: one joint turns the whole arm horizontally, and the other two move it up and down in a vertical plane. All the links are 3D printed, and each joint is driven directly by its servo with no gearbox.

Joint	Servo / pin	Axis	What it does
J1: base	Servo 1 · D3	Vertical (yaw)	Turns the arm between the pick area and the drop area
J2: elbow	Servo 2 · D5	Horizontal (pitch)	Raises and lowers the arm, setting how far it reaches and how high the magnet is
J3: wrist	Servo 3 · D6	Horizontal (pitch)	Points the electromagnet down at the material
End effector	Relay	—	The electromagnet, switched on to pick up and off to release

Kinematics (where the magnet ends up)

With joint angles θ_1 (base), θ_2 (elbow) and θ_3 (wrist), link lengths L_1 (elbow to wrist) and L_2 (wrist to magnet tip), and pivot height h, the electromagnet's position is:

```
r = L1·cos θ2 + L2·cos(θ2 + θ3)          horizontal reach
z = h + L1·sin θ2 + L2·sin(θ2 + θ3)      height of the magnet tip
x = r·cos θ1 ,  y = r·sin θ1             position on the work surface
```

The 2023 build was calibrated by trial and error: each target angle was adjusted until the magnet sat over the material. With these equations, the pick and drop positions can instead be calculated from measured coordinates, which is how the v2 design (Section 9) defines its poses.

Mechanical considerations

- **Torque at the base of the arm:** the elbow servo carries the arm and the electromagnet with the maximum lever arm when the arm is stretched out horizontally. That makes it the most heavily loaded joint, which is why metal-gear 11 kg·cm MG996Rs were chosen over plastic-gear micro servos.
- **Light links:** 3D-printed links keep the arm's own weight low, leaving more of the servo torque for the payload.
- **Electromagnet at the end of the arm:** a coil wound directly round the end of the arm adds very little weight, compared with a mechanical gripper and its own actuator.

4 Electromagnet and relay

How the magnet works

Current through a coil wound on a soft-iron core produces a magnetic field. The core concentrates the field, so the coil is many times stronger than it would be in air. For a long coil:

```
B ≈ μ0 · μr · N · I / l
F ≈ B² · A / (2 · μ0)       (pull on a part touching the core face)
```

N is the number of turns, I the current, l the length of the coil, and A the area of the core's face. The design follows from these equations:

- **Strength depends on current:** more turns or more current give a stronger magnet, up to the point where the iron core saturates. Unlike a gripper, its lifting capacity depends on the coil current, not on servo torque.
- **Contact matters:** the force grows with B^2, and B drops quickly across even a small air gap. So the arm's path lowers the magnet right down onto the material.
- **Release:** turning the current off removes almost all the field. A small leftover magnetism in the core can hold very fine filings; v2 deals with this (Section 9).

Why a relay

An Arduino pin can supply at most about 40 mA (20 mA continuous), while the coil needs far more current than that. The relay solves this:

- **Isolation:** a small control current operates the relay contact, which switches the coil's current from a separate battery. The coil current never passes through the microcontroller.
- **Separate power:** the coil runs from the Li-ion pack, protected by its BMS, so magnet current surges can't reset the Arduino or disturb the servo signals.
- **Simple control:** the magnet is just on or off, which is easy to read, test and demonstrate.

> **In the 2023 demo** the relay, and so the electromagnet, was switched **by hand** at the pick and drop points, while the Arduino ran the arm's movement. Connecting the relay to the Arduino so the whole cycle runs on its own is the main change in v2.

Integration challenge

Getting the relay, battery and electromagnet to work together was the main technical challenge of the project, and it was my part of the work. It involved four things:

- sizing the winding so the magnet reliably lifted the test material;
- giving the coil its own power source so it didn't disturb the logic supply;
- wiring the relay's contact side for the coil current;
- making soldered joints that survive the arm's repeated movement, because loose jumper wires came off during motion.

5 Firmware (2023 version)

I wrote and calibrated the firmware. The sketch uses the Arduino `Servo` library, which generates the 50 Hz servo pulses from a hardware timer. It plays back a fixed sequence of joint movements, and moves each joint **one degree every 20 ms** instead of jumping straight to the target angle.

```
#include <Servo.h>
Servo s1, s2, s3;           // base, elbow, wrist
int pos1, pos2, pos3;

void setup() {
  s1.attach(3);             // PWM pins
  s2.attach(5);
  s3.attach(6);
}

void loop() {
  delay(500);
  s1.write(90); delay(100);  // home pose
  s2.write(90); s3.write(90);

  for (pos1 = 90; pos1 ⩾ 0; pos1--)  // rotate to pick side
    { s1.write(pos1); delay(20); }
  delay(1000);
  for (pos2 = 90; pos2 ⩾ 20; pos2--) // lower the elbow
    { s2.write(pos2); delay(20); }
  delay(1000);
  for (pos3 = 90; pos3 ⩽ 140; pos3++) // tilt the wrist down
    { s3.write(pos3); delay(20); }
  delay(1000);
  for (pos3 = 140; pos3 ⩽ 100; pos3--) // (see Section 8)
    { s3.write(pos3); delay(20); }
  delay(1000);
  for (pos2 = 20; pos2 ⩽ 45; pos2++)  // lift
    { s2.write(pos2); delay(20); }
  delay(1000);
  for (pos1 = 0; pos1 ⩽ 90; pos1++)  // rotate back
    { s1.write(pos1); delay(20); }
  delay(2000);
}
```

Motion sequence

Step	Joint	Angle	Time
Home	all	90° / 90° / 90°	0.6 s
Rotate	Base	90° → 0°	1.8 s
Lower	Elbow	90° → 20°	1.4 s
Tilt	Wrist	90° → 140°	1.0 s
Lift	Elbow	20° → 45°	0.5 s
Return	Base	0° → 90°	1.8 s
Pauses	—	5 × 1 s + 2 s	7.0 s
Total per cycle			**≈ 14.1 s**

WHY MOVE ONE DEGREE AT A TIME

- Moving 1° every 20 ms caps each joint at 50 °/s. That is well below the MG996R's maximum of about 315 °/s at 6 V, so the arm and the magnet's load move smoothly and don't jerk.
- Slow movement also keeps the current each servo draws low, so the servos don't pull the supply voltage down and brown out the board.
- Twenty milliseconds is one servo PWM period, so every step gets exactly one new pulse.

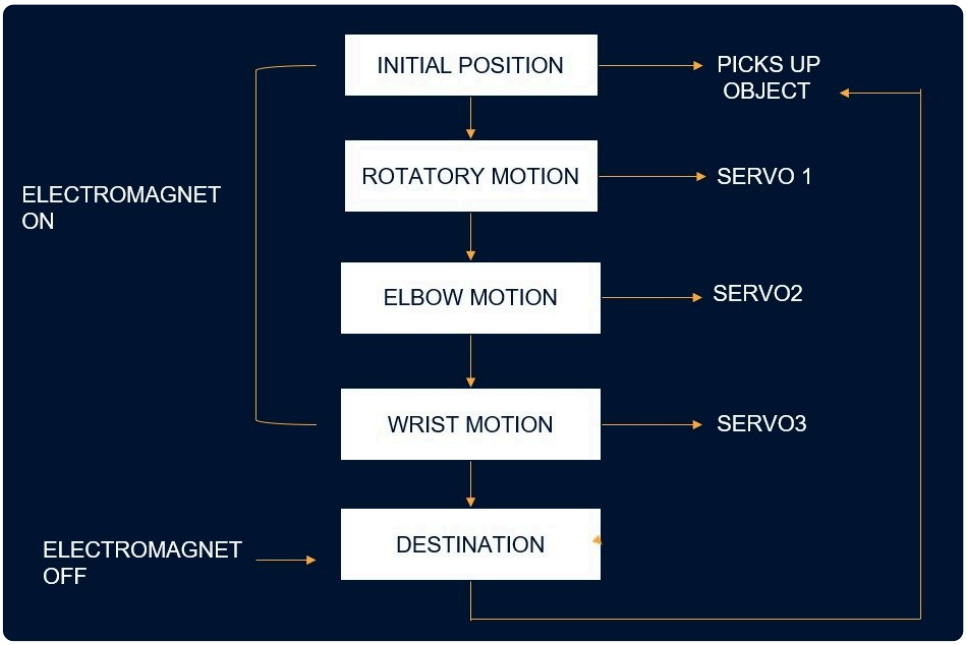


Figure 4. Original 2023 operating flow: initial position → rotation (servo 1) → elbow (servo 2) → wrist (servo 3) → destination, with the electromagnet on while carrying and off at the destination.

6 Testing and results

The arm was tested in two hands-on tests, each repeated, and watched for whether the joints reached their programmed angles and whether the magnet picked up and released the material.

Test	Procedure	Result
T1: iron filings out of a mixture	Iron filings mixed with non-magnetic material. The arm lowered the energised magnet onto the mixture, lifted, rotated to the drop side, and the magnet was switched off.	Passed The filings were lifted out and the non-magnetic material was left behind.
T2: move a nail to a target spot	A small iron nail was picked up at the pick position and released at a set drop position.	Passed The nail was picked up and dropped at the set spot.
T3: joint angles	Each servo's movement was checked against its programmed angle range over repeated cycles.	Passed Each joint moved through its programmed range.
T4: relay switching	The relay was switched repeatedly with the coil on battery power.	Passed The magnet switched on and off cleanly.

These were qualitative checks on a working prototype. Payload, positioning accuracy and cycle-to-cycle repeatability were **not** measured in 2023. The v2 test plan in Section 9 adds those measurements.

Applications

SCRAP SORTING

Pulling ferrous scrap out of mixed waste, a scaled-down version of what magnetic separators do in steel recycling.

SMALL-PART HANDLING

Moving screws, nails and small steel parts during assembly, or between containers, without a gripper that has to fit each part.

HAZARDOUS HANDLING

Collecting sharp metal debris so nobody has to handle it, the original motivation for the project.

Limitations of the 2023 build

- It only picks up ferromagnetic material, and the coil needs current the whole time it is holding something.
- It is a small, desktop-sized prototype; the payload is limited by elbow torque and the strength of the coil.
- The movement is fixed in advance: there is no sensing, so the material has to be in the expected place.

7 What I learned

ENGINEERING

- Keeping high-current loads (coil, servos) on separate power from the logic circuit.
- Servo PWM timing, and how the speed of a movement affects how much current it draws.
- Choosing actuators by the torque needed at the most heavily loaded joint.
- Electromagnet design: number of turns, current, core saturation, air gap.

PROCESS

- Going from requirements through CAD, 3D printing, assembly, programming and testing, to a working demo by a fixed deadline.
- Writing it up for assessment: report, poster, slides and a live demo.
- Integrating my electronics and firmware with the mechanical build of the arm.

8 2026 engineering review

This review and the v2 design are my own independent work, done in 2026. I went back over the 2023 build with three more years of embedded experience. The prototype met its goal, but a review of its code and wiring shows what I'd change before calling it reliable.

#	Finding	Effect	Fix in v2
1	**Loop that never runs:** `for (pos3 = 140; pos3 ⩽ 100; pos3--)`	The condition is false from the start (140 is not ≤ 100), so the wrist's return-to-100° step never happens. The wrist stays at 140°.	A single `moveTo()` function that works out the direction itself
2	**Joints jump back to home** at the start of each cycle	The next pass of `loop()` writes 90° straight away while the elbow is at 45° and the wrist at 140°. They jump at full speed: a jerk in the arm and a current spike.	Return to home smoothly, and set the starting angle before `attach()`
3	**No relay control in the code**	The magnet had to be switched by hand, so the arm couldn't run a cycle on its own and the on/off timing varied between runs.	Relay on D7, switched at the right point in the sequence
4	**Joints move one after another**	Only one joint moves at a time, which makes the cycle longer (≈ 14 s) than it needs to be.	Interpolate all joints together so they arrive at the same moment
5	**Angles typed into the code by hand**	Changing a pick or drop location means editing several loops.	A `Pose` table, with values calculated from the kinematics
6	**No flyback diode** on the coil	When the relay opens, the coil's collapsing field produces a voltage spike that wears the relay contacts and can cause interference.	Flyback diode (1N4007, or a Schottky diode) across the coil
7	**How the servos are powered**	Three MG996Rs can each draw up to 2.5 A when stalled. They need a dedicated 4.8–6 V supply with a ground shared with the Arduino, not the Arduino's 5 V pin.	A 6 V, 5 A step-down regulator for the servos, plus a bulk capacitor
8	**Nothing measured**	No payload, accuracy or repeatability numbers.	Test plan with logged measurements (Section 9)

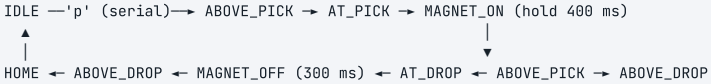
9 v2 design: fully automatic cycle Proposed

The redesign keeps the same arm and fixes every item in the review. It has not been built; the pose angles in the code are placeholders to calibrate on the real arm.

Hardware changes

- **Relay controlled by the Arduino on D7** (some relay modules are active-LOW, so the code has a setting to invert the signal). An even better option: a logic-level N-MOSFET instead of the relay. It switches silently with no contact wear, and driven with PWM it can set the magnet strength and briefly reverse polarity to release filings that stay stuck by leftover magnetism.
- **Flyback diode** across the electromagnet coil.
- **6 V step-down regulator for the servos** with a 1000 µF bulk capacitor and a ground shared with the Arduino. The servos never draw current from the Arduino's 5 V pin.
- **Limit or home switch** on the base, so the arm can find its reference position on power-up.

Control flow



v2 test plan

Metric	Method
Payload	Add steel washers until the part slips; log the coil current
Repeatability	Mark where the tip lands over 20 cycles; measure the spread
Separation rate	Weigh the filings recovered from a known mixture
Cycle time	Timestamp each step over serial

```
#include <Servo.h>

const uint8_t PIN_BASE = 3, PIN_ELBOW = 5,
              PIN_WRIST = 6, PIN_MAGNET = 7;
const bool    RELAY_ACTIVE_HIGH = true;
const uint8_t STEP_MS = 15;          // ~67 deg/s
const uint16_t SETTLE_MS = 300;

struct Pose { uint8_t base, elbow, wrist; };
const Pose HOME       = { 90, 90,  90 };
const Pose ABOVE_PICK = {  0, 60, 120 };  // calibrate
const Pose AT_PICK    = {  0, 20, 140 };
const Pose ABOVE_DROP = {150, 60, 120 };
const Pose AT_DROP    = {150, 45, 110 };

Servo sBase, sElbow, sWrist;
Pose cur = HOME;

// All joints move together and arrive at once
void moveTo(const Pose &t) {
  int db = t.base  - cur.base;
  int de = t.elbow - cur.elbow;
  int dw = t.wrist - cur.wrist;
  int n = max(abs(db), max(abs(de), abs(dw)));
  for (int i = 1; i <= n; i++) {
    sBase.write (cur.base  + (long)db * i / n);
    sElbow.write(cur.elbow + (long)de * i / n);
    sWrist.write(cur.wrist + (long)dw * i / n);
    delay(STEP_MS);
  }
  cur = t;
  delay(SETTLE_MS);
}

void magnet(bool on) {
  digitalWrite(PIN_MAGNET,
               on == RELAY_ACTIVE_HIGH ? HIGH : LOW);
}

void setup() {
  pinMode(PIN_MAGNET, OUTPUT);
  magnet(false);
  // write before attach: no jump at power-on
  sBase.write(HOME.base);   sBase.attach(PIN_BASE);
  sElbow.write(HOME.elbow); sElbow.attach(PIN_ELBOW);
  sWrist.write(HOME.wrist); sWrist.attach(PIN_WRIST);
  Serial.begin(115200);
  Serial.println(F("ERA v2: send 'p' to run one cycle"));
}

void pickAndPlace() {
  moveTo(ABOVE_PICK); moveTo(AT_PICK);
  magnet(true);  delay(400);
  moveTo(ABOVE_PICK); moveTo(ABOVE_DROP); moveTo(AT_DROP);
  magnet(false); delay(300);
  moveTo(ABOVE_DROP); moveTo(HOME);
}

void loop() {
  if (Serial.available() && Serial.read() == 'p')
    pickAndPlace();
}
```

10 Future work

WIRELESS CONTROL

Move to an ESP32 (as the 2023 report proposed with a NodeMCU) so the arm can be controlled and monitored from a phone over BLE or Wi-Fi.

SEEING THE MATERIAL

Add a camera, or an inductive proximity sensor, to find ferrous material and calculate the pick pose with the kinematics, instead of always going to a fixed spot.

POSITION FEEDBACK

Servos with position feedback, or an angle sensor on each joint, for closed-loop position control and stall detection.

Built in 2023 as part of a 4-member academic mini-project team under the guidance of Prof. Chandra Shekar N, Dept. of ECE, DSCE, Bengaluru. Mechanical design adapted from an open-source desktop robotic-arm model. Sections 8–10 are the author's independent 2026 work and describe a proposed design that has not been built.